Haniyeh Ehsani Oskouie

haniyeh@cs.ucla.edu • Website • Google Scholar • LinkedIn • GitHub

Summary

PhD candidate at UCLA researching **robust and efficient adaptation of large language/vision language models** with work spanning **robustness**, **parameter-efficient fine-tuning and post-training** , and **trustworthy evaluation and control (benchmarking)**. Published research at **NeurIPS, ECCV, ICML, and ICLR**, with industry experience at **Pinterest Labs**, **Scale AI**, and **Handshake**.

Education

- **University of California, Los Angeles (UCLA)**
  Computer Science (Doctorate of Philosophy) *2023-Expected 2028*
  GPA: 3.82/4 | Advisor: Prof. Majid Sarrafzadeh
- **University of California, Los Angeles (UCLA)**
  Computer Science - Artificial Intelligence (Master of Science, en route to Ph.D.) *2023-2025*
  GPA: 3.82/4 | Advisor: Prof. Majid Sarrafzadeh
- **Sharif University of Technology**
  Computer Engineering (Bachelor of Science) *2018–2023*
  GPA (154 Total Credits): 18.24/20 (3.72/4) | Advisor: Prof. Shohreh Kasaei

Related Courses

Computational Ethics of Natural Language Processing, Artificial Life for Computer Graphics & Vision, Reinforcement Learning, Computer Animation, Advanced Data Mining, Large-scale Machine Learning, Advanced Neural Networks & Deep Learning, Machine Learning, Artificial Intelligence, Modern Information Retrieval, Image Processing

Experience

- **Instructor/Research Assistant at the University of California, Los Angeles**
  UCLA Vision Lab, eHealth Research Lab *Sep 2023 - Present*
  Developed parameter-efficient learning methods for vision-language models and optical flow, alongside research on adversarial robustness, sample-efficiency estimation, and LLM-based toxicity classification, resulting in publications at NeurIPS 2026, ECCV 2026, ICLR 2026, ICAD 2026, and ICNC 2025.
- **Research Intern at Pinterest Labs**
  Ads Creative *Jun 2026 - Sep 2026*
  Optimized an agentic AI pipeline for ad creative generation that pairs diffusion models for product background generation with VLM judges, fine-tuned via LoRA-based SFT and DPO on human reviewed data, for automated safety assessment, achieving a 41.97% improvement in defect detection accuracy. Paper currently in preparation.
- **Research Intern at Scale AI + Outlier**
  Human Frontier Collective *May 2025 - Sep 2026*
  Built and evaluated LLM benchmarking datasets for scientific reasoning tasks, including SciPredict, by designing data-quality checks, evaluating model calibration under knowledge and feasibility constraints, and analyzing failure modes across 405 tasks and 33 scientific domains; co-authored a resulting ICML 2026 / ICLR workshop 2026 publication.
- **Machine Learning Expert at Handshake AI**
  AI Trainer (Joint project with OpenAI) *Jul 2025 - Sep 2025*
  Following the PaperBench pipeline, converted 9 ICLR papers into runnable PyTorch code, designed evaluation rubrics to assess correctness, and graded LLM-generated implementations.
- **Research Intern at Stealth Startup**
  Computational Psychiatry Lab *Jun 2024 - Aug 2024*
  Performed pose estimation through video processing and frequency-domain denoising for exercise posture correction, and investigated movement- and sensor-based indicators of mental well-being. Improved physiological measurement quality by 10.23%.
- **Research Intern at Iran Telecommunication Research Center (Joint with my Bachelor's Thesis)**
  AI Labs *Oct 2022 - Feb 2023*
  Designed and evaluated FGSM-SF/PGD-SF white-box adversarial attacks for point-cloud scene-flow networks, inducing up to 33.7% relative AEPE degradation, and analyzed sensitivity to single-dimension/color-channel perturbations (MLSP 2024).

- **Research Intern at the University of Birmingham**
  Cognitive Robots and Systems Lab, Under the supervision of Prof. Mohan Sridharan *Mar 2022 - Apr 2023*
  Contributed to designing a hybrid ad hoc teamwork architecture that combines non-monotonic logical reasoning with commonsense knowledge and rapidly learned teammate-behavior models from limited examples (AAAI 2023).
- **Research Intern at the Chinese University of Hong Kong**
  Under the supervision of Prof. Farzan Farnia *Jul 2022 - Oct 2022*
  Demonstrated that gradient-based saliency methods are vulnerable to universal adversarial perturbations, increasing explanation dissimilarity scores by up to $1.5\times$, and developed Universal Perturbations for Interpretation using gradient- and PCA-based optimization to reliably manipulate model explanations (ICASSP 2023).

For more details kindly refer to my website.

Selected Publications

- D. Lao, Y. Zhang, **H. Ehsani Oskouie**, Y. Wu, A. Wong, S. Soatto, "Test-Time Defense Against Adversarial Attack by Stochastic Resonance," *NeurIPS 2026.*
- S. Ghiasvand, **H. Ehsani Oskouie**, M. Alizadeh, R. Pedarsani, "MMLoP: Multi-Modal Low-Rank Prompting for Efficient Vision-Language Adaptation," *ECCV 2026.*
- Y. Xiao, C. R. Coronel, J. Zhan, **H. Ehsani Oskouie**, A. Wong, D. Lao, "Triangular Consistency as a Universal Constraint for Optical Flow Learning," *ECCV 2026.*
- U. M. Sehwag, E. Lau, **H. Ehsani Oskouie**, S. Shabihi, E. Liang, A. Toledo, G. Mangialardi, S. Fonrouge, E. H. Cardona, P. Vergara, U. Tyagi, C. B. C. Zhang, P. Bhatter, N. Johnson, F. Huang, E. G. H. Montoya, and B. Liu, "SciPredict: Can LLMs Predict the Outcomes of Research Experiments in Natural Sciences?" *ICML 2026, FM4Science @ ICLR 2026.*
- **H. Ehsani Oskouie***, S. Ghiasvand*, M. Alizadeh, R. Pedarsani, "Few-Shot Adversarial Low-Rank Fine-Tuning of Vision-Language Models," *ICLR 2026.*
- A. Hatamian, **H. Ehsani Oskouie***, L. Levine*, M. Sarrafzadeh, "Exploring the Impact of Dataset Statistical Effect Size on Model Performance and Data Sample Size Sufficiency," *IEEE ICAD 2026.*
- **H. Ehsani Oskouie***, C. Chance*, C. Huang*, M. Capetz*, E. Eyeson*, M. Sarrafzadeh, "Leveraging Large Language Models and Topic Modeling for Toxicity Classification," *IEEE ICNC 2025.*
- **H. Ehsani Oskouie**, MS. Moin, and S. Kasaei, "Attack on Scene Flow using 3D Point Clouds," *IEEE MLSP 2024.*
- **H. Ehsani Oskouie**, and F. Farnia, "Interpretation of Neural Networks is Susceptible to Universal Adversarial Perturbations," *IEEE ICASSP 2023.*
- S. Ghiasvand, M. Amirizaniani, **H. Ehsani Oskouie**, M. Alizadeh, R. Pedarsani, "Can MLLMs Critique Like Humans? Evaluating Open-Ended Aesthetic Reasoning in Multimodal Large Language Models," under review at *WACV 2027.*
- S. Ghiasvand, **H. Ehsani Oskouie**, S. Mansouri, M. Alizadeh, F. Farnia, R. Pedarsani, "Kernel-Based Steering of CLIP with Vision-Language Model Preferences," under review at *ICLR 2027.*

* contributed equally.

Technical Skills

- **Machine Learning Tools:** PyTorch, Tensorflow, Hugging Face, OpenCV, timm, torchvision, Pandas, Numpy, Scikit-learn, Matplotlib, Matlab
- **Programming Languages:** Python, Java, JavaScript, C, R, MySQL
- **Tools/Systems:** Git, Linux, Docker, Jupyter, Weights & Biases

Honors and Awards

- **Reviewer**: **PLOS ONE 2025**, **TMLR 2026**, and **WCCI 2026**.
- **Fellowships**: **Human Frontier Collective Fellowship 2025-2026** (Scale AI); **MOVE Fellowship 2025** (Handshake AI).
- Ranked **top 0.1%** in the National Universities Entrance Exam for Mathematics and Physics Majors among over 140,000 participants.
- **Robotics**: Placed **1st in 2015** and **2nd/3rd/4th in 2016** in national robotics competitions.